# Biology 101 — Chapter 4: Cell Structure and Energy

## 1. The Cell Membrane

The cell membrane, also known as the plasma membrane, is a biological membrane that separates the interior of the cell from the outside environment. It is composed primarily of a phospholipid bilayer embedded with proteins, cholesterol, and carbohydrates. The membrane is selectively permeable, meaning it controls what enters and leaves the cell, maintaining homeostasis and protecting the cell from its surroundings.

The fluid mosaic model describes the cell membrane as a dynamic structure in which proteins and lipids can move laterally within the plane of the membrane. This fluidity is essential for processes such as cell signaling, endocytosis, and membrane trafficking. The ratio of saturated to unsaturated fatty acids affects membrane fluidity, with unsaturated fats increasing fluidity at lower temperatures.

Transport across the membrane occurs through passive and active mechanisms. Diffusion and facilitated diffusion move substances down their concentration gradient without energy input, while active transport uses ATP to move molecules against their gradient. The sodium-potassium pump is a classic example of active transport, exporting three sodium ions and importing two potassium ions per ATP consumed.

## 2. Mitochondria and Energy

Mitochondria are double-membraned organelles often called the powerhouse of the cell because they generate most of the cell's supply of adenosine triphosphate, or ATP, which is used as a source of chemical energy. The inner membrane is folded into structures called cristae, which increase the surface area available for oxidative phosphorylation.

Cellular respiration occurs in three main stages: glycolysis, the citric acid cycle, and oxidative phosphorylation. Glycolysis takes place in the cytoplasm and breaks one molecule of glucose into two molecules of pyruvate, producing a small net gain of two ATP molecules. The citric acid cycle then oxidizes pyruvate derivatives in the mitochondrial matrix, releasing carbon dioxide and transferring electrons to carrier molecules.

The electron transport chain, located in the inner mitochondrial membrane, uses these carriers to pump protons into the intermembrane space, creating a proton gradient. ATP synthase then harnesses this gradient to phosphorylate ADP, producing approximately thirty molecules of ATP per glucose molecule. This process requires oxygen as the final electron acceptor, which is why aerobic organisms depend on respiration.

## 3. Photosynthesis Overview

Photosynthesis is the process by which green plants, algae, and some bacteria convert light energy into chemical energy stored in glucose. It takes place primarily in the chloroplast, an organelle containing the pigment chlorophyll, which absorbs light most strongly in the blue and red regions of the spectrum while reflecting green light.

The light-dependent reactions occur in the thylakoid membranes, where photosystems capture photons and use the energy to split water molecules, release oxygen, and produce ATP and NADPH. The Calvin cycle, which takes place in the stroma, then uses these products to fix carbon dioxide into three-carbon sugars that can be combined into glucose and other carbohydrates.

The overall equation for photosynthesis combines six molecules of carbon dioxide and six molecules of water to produce one molecule of glucose and six molecules of oxygen in the presence of light. Photosynthesis is fundamental to life on Earth because it produces the oxygen in our atmosphere and forms the base of nearly every food chain.